Chapter 6 - The Riemann Integral

Calculus I

November 20, 2024

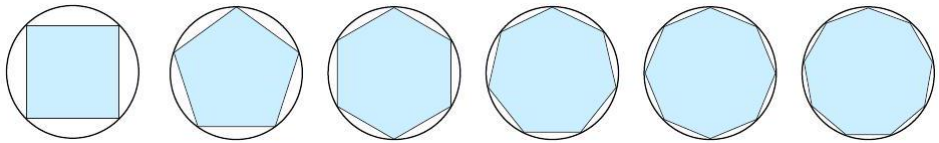
Table of Contents

1. Examples of Riemann Integrals
2. Definition and Properties of Riemann Integral
3. Interpretations of Riemann Integral
4. The Fundamental Theorem of Calculus
5. Integration by Parts and Change of Variables
6. Improper Integrals

Section 1

Examples of Riemann Integrals

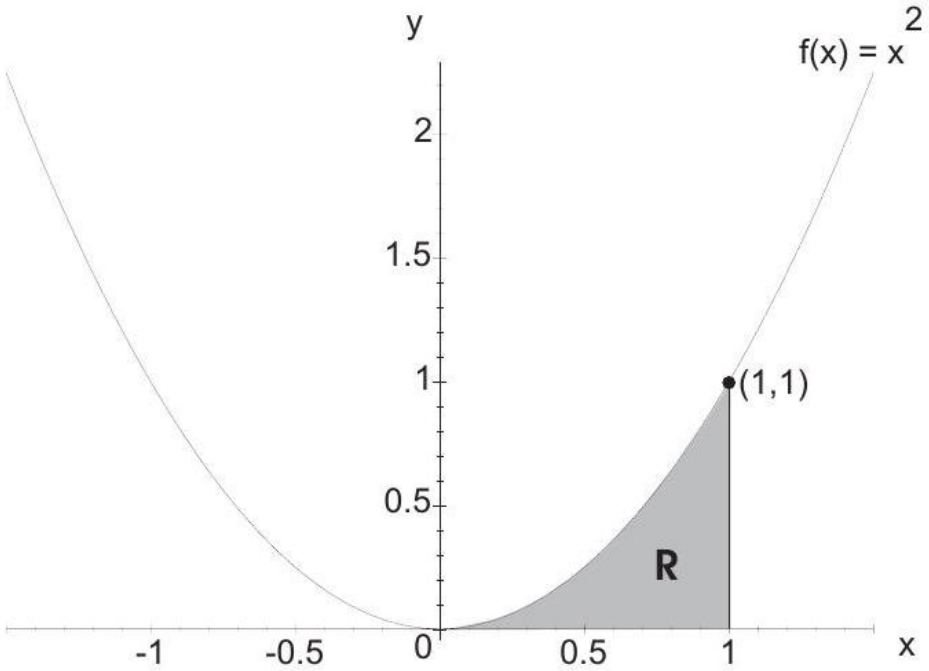
Area of a Circle



Question: How do you find the area of a circle?

Historical Note: Eudoxus of Cnidus and Archimedes used a type of limit process known as the the Method of Exhaustion to calculate areas.

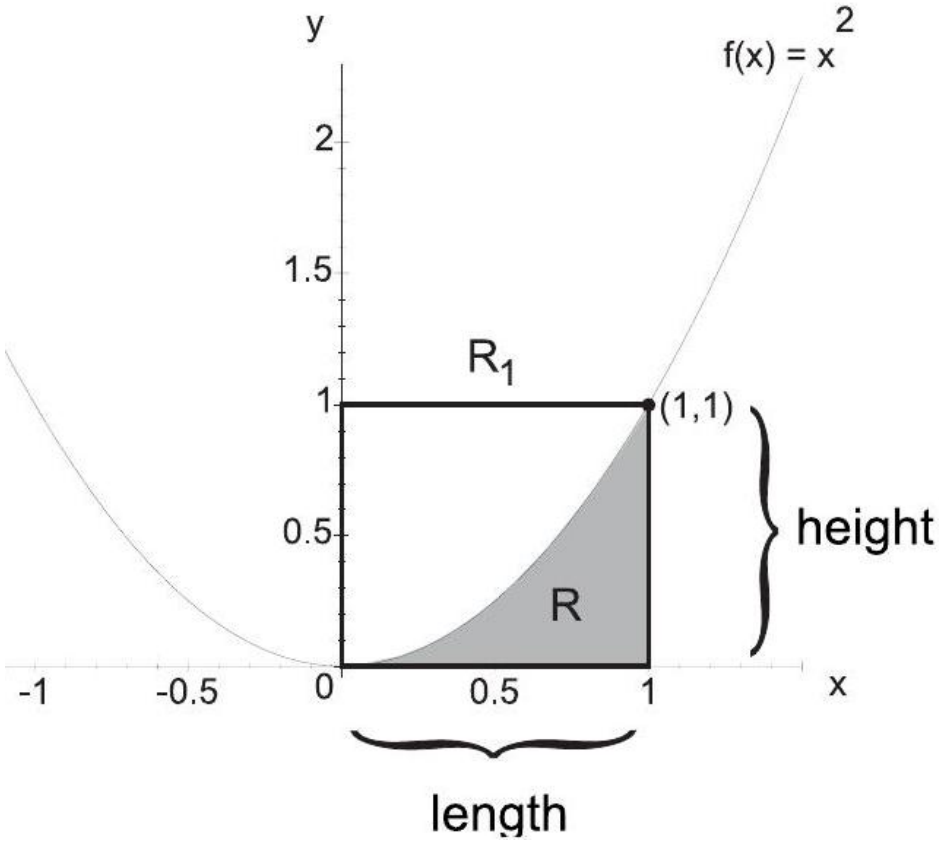
Area Under Curves



Problem: Suppose that $f(x) = x^2$. Consider the region R bounded by the graph of f, by the x-axis, and by the lines $x = 0$ and $x = 1$.

How could we determine the area of this irregular region?

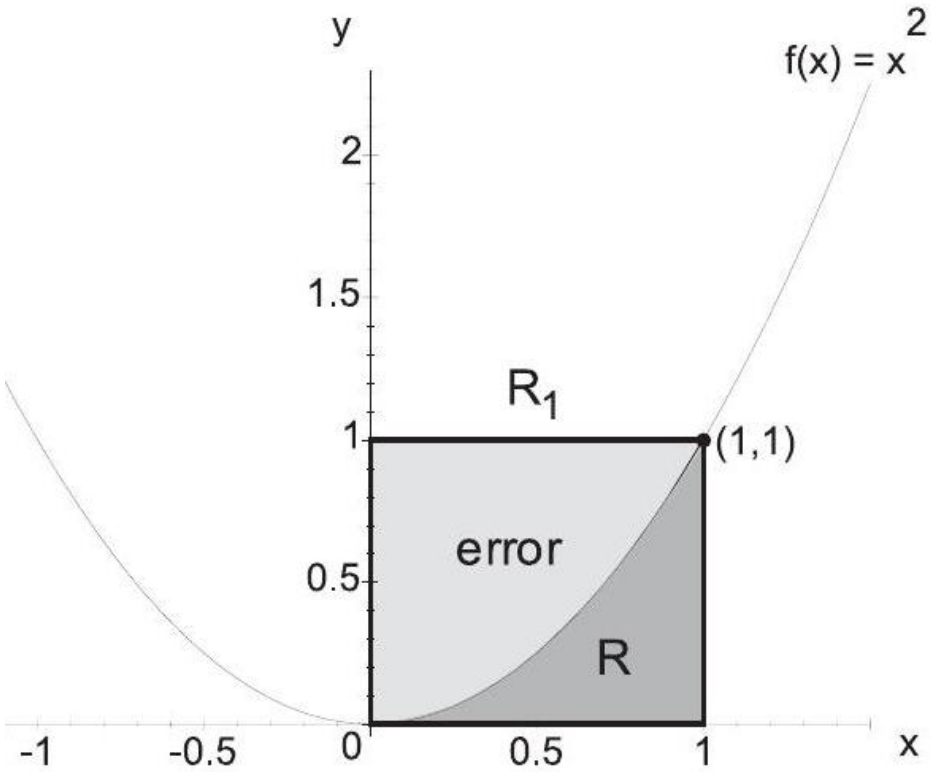
Area Under Curves



$$\text{Estimate} = \boldsymbol{R}_1$$

$$= \text{ length } \times \text{ height } = 1 \times f(1) = 1 \times 1^2 = 1$$

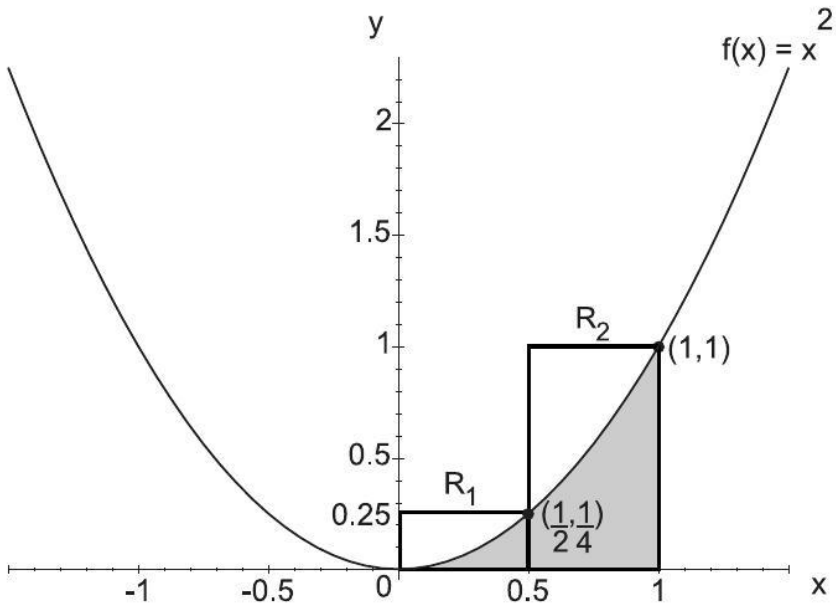
Area Under Curves



$$\text{Estimate} = \boldsymbol{R}_1$$

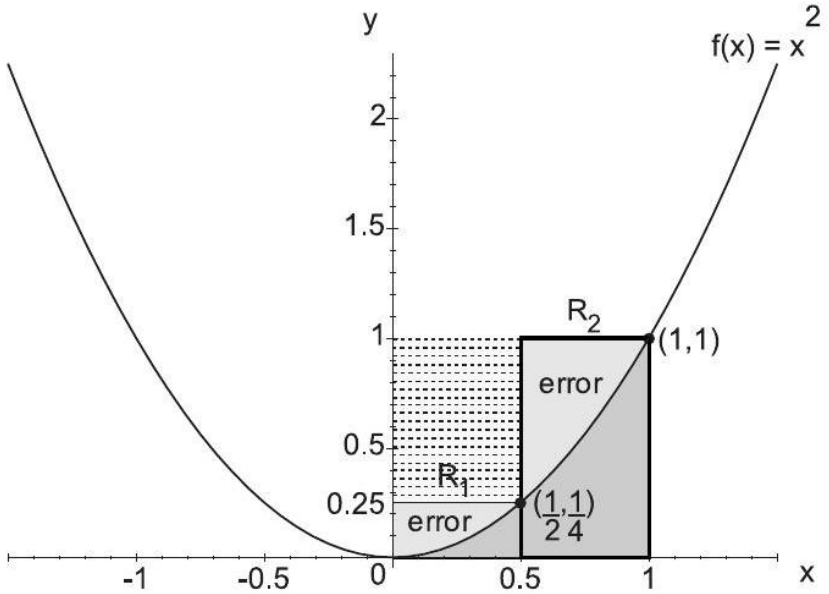
$$= \text{ length } \times \text{ height } = 1 \times f(1) = 1 \times 1^2 = 1$$

Area Under Curves



$$\begin{aligned}
\text{Estimate } = R_1 + R_2 &= \left(\frac{1}{2} \times f\left(\frac{1}{2}\right)\right) + \left(\frac{1}{2} \times f(1)\right) \\
&= \left(\frac{1}{2} \times \left(\frac{1}{2}\right)^2\right) + \left(\frac{1}{2} \times 1^2\right) \\
&= \left(\frac{1}{2} \times \frac{1^2}{2^2}\right) + \left(\frac{1}{2} \times \frac{2^2}{2^2}\right) \\
&= 0.625
\end{aligned}$$

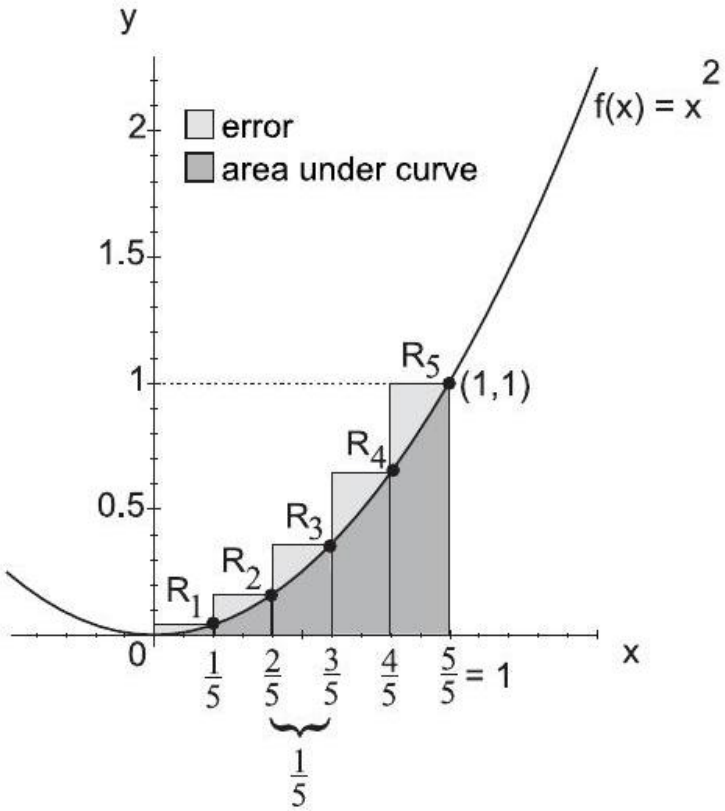
Area Under Curves



$$\begin{aligned}
\text{Estimate } = R_1 + R_2 &= \left(\frac{1}{2} \times f\left(\frac{1}{2}\right)\right) + \left(\frac{1}{2} \times f(1)\right) \\
&= \left(\frac{1}{2} \times \left(\frac{1}{2}\right)^2\right) + \left(\frac{1}{2} \times 1^2\right) \\
&= \left(\frac{1}{2} \times \frac{1^2}{2^2}\right) + \left(\frac{1}{2} \times \frac{2^2}{2^2}\right) \\
&= 0.625
\end{aligned}$$

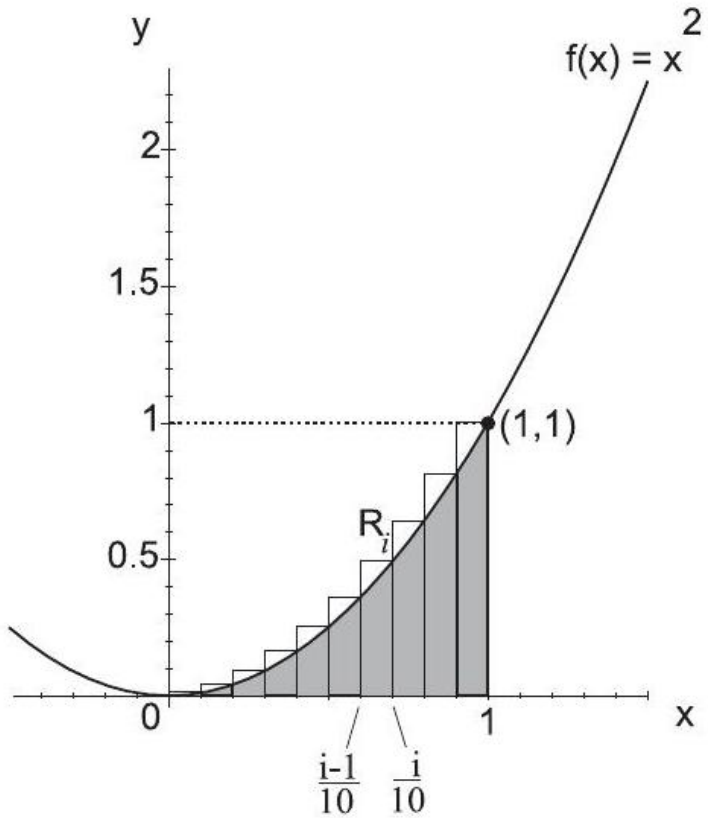
Area Under Curves

Estimate $= R_1 + R_2 + R_3 + R_4 + R_5$



$$
\begin{aligned}
&= (\frac{1}{5} \times f(\frac{1}{5})) + (\frac{1}{5} \times f(\frac{2}{5})) + (\frac{1}{5} \times f(\frac{3}{5})) + (\frac{1}{5} \times f(\frac{4}{5})) + (\frac{1}{5} \times f(\frac{5}{5})) \\
&= (\frac{1}{5} \times \frac{1^2}{5^2}) + (\frac{1}{5} \times \frac{2^2}{5^2}) + (\frac{1}{5} \times \frac{3^2}{5^2}) + (\frac{1}{5} \times \frac{4^2}{5^2}) + (\frac{1}{5} \times \frac{5^2}{5^2}) \\
&= \frac{1}{5^3} \times (1^2 + 2^2 + 3^2 + 4^2 + 5^2) \\
&= 0.44
\end{aligned}
$$

Area Under Curves



$$\sum_{i=1}^{10} R_i = \sum_{i=1}^{10} \frac{1}{10} \times \frac{i^2}{10^2}$$

$$= \frac{1}{10^3} \times \sum_{i=1}^{10} i^2$$

$$= \frac{1}{10^3} \times \frac{(10)(10+1)(2(10)+1)}{6}$$

$$= 0.385$$

Area Under Curves

With 1000 subintervals the area estimate is:

$$\begin{aligned}\sum_{i=1}^{1000} R_i &= \sum_{i=1}^{1000} \frac{1}{1000} \times \frac{i^2}{1000^2} \\ &= \frac{1}{1000^3} \times \sum_{i=1}^{1000} i^2 \\ &= \frac{1}{1000^3} \times \frac{(1000)(1000+1)(2(1000)+1)}{6} \\ &= 0.3338335\end{aligned}$$

Area Under Curves

With n subdivisions the area estimate is:

$$\begin{aligned}\sum_{i=1}^{n} R_i &= \frac{1}{n^3}\sum_{i=1}^{n} i^2 \\ &= \frac{1}{n^3}\frac{(n)(n+1)(2(n)+1)}{6} \\ &= \frac{\frac{1}{n^3}\left(2n^3+3n^2+n\right)}{6} \\ &= \frac{2+\frac{3}{n}+\frac{1}{n^2}}{6}\end{aligned}$$

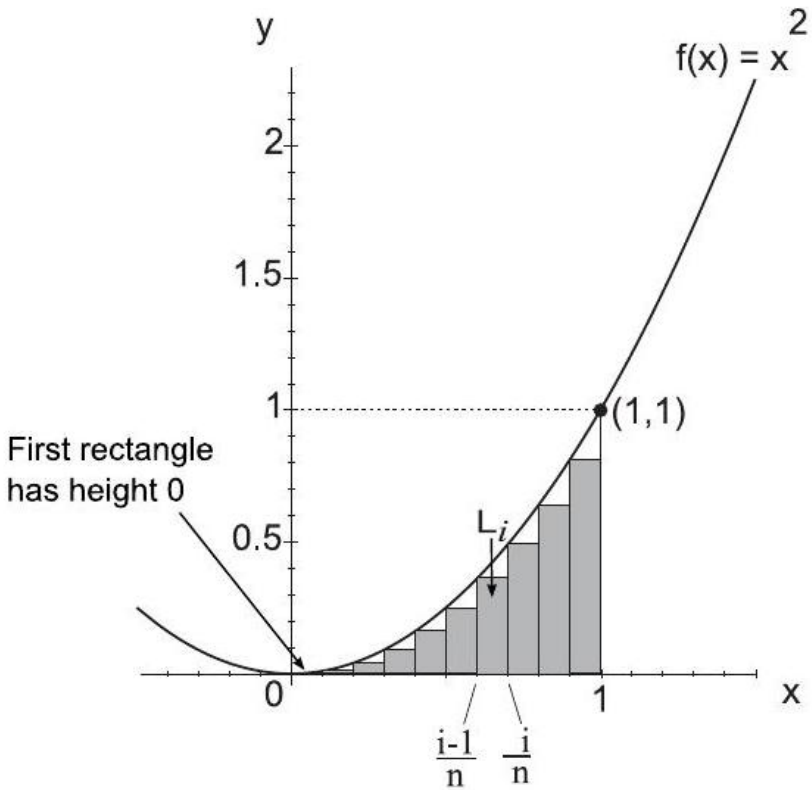
Key Observation: We have

$$\lim_{n\to\infty}\sum_{i=1}^{n} R_i = \lim_{n\to\infty}\frac{2+\frac{3}{n}+\frac{1}{n^2}}{6} = \frac{1}{3}$$

Area Under Curves

Number of Subintervals (Rectangles)	Length of Subinterval (Width of Rectangle)	Estimate for Area under Curve
1	1	1
2	$\frac{1}{2}$	0.625
5	$\frac{1}{5}$	0.44
10	$\frac{1}{10}$	0.385
1000	$\frac{1}{1000}$	0.3338335
approaches ∞	approaches 0	approaches $\frac{1}{3}$

Area Under Curves



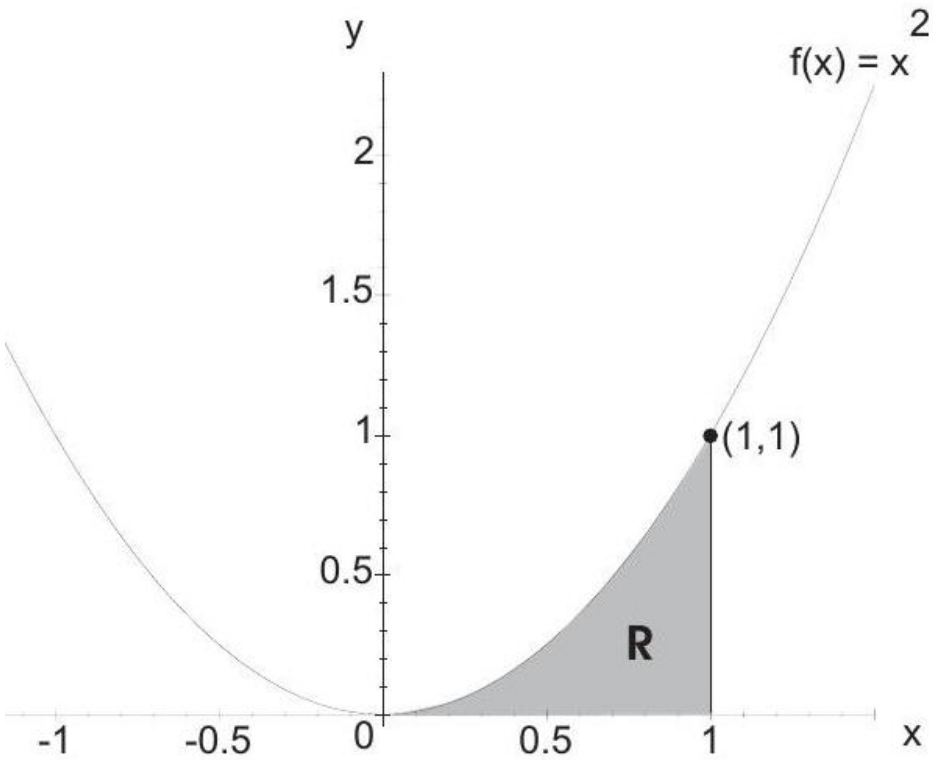
$$\sum_{i=1}^{n} L_i = \frac{1}{n^3}\sum_{i=1}^{n}(i-1)^2$$

$$= \frac{1}{n^3}\frac{(n-1)(n+1-1)(2(n-1)+1)}{6}$$

$$= \frac{1}{n^3}\left(\frac{2n^3-3n^2+n}{6}\right)$$

$$= \frac{2-\frac{3}{n}+\frac{1}{n^2}}{6} \to \frac{1}{3}$$

Area Under Curves



Conclusion:

$$\text{Area} = \frac{1}{3}$$

Section 2

Definition and Properties of Riemann Integral

Introduction to Riemann Sums

Definition 6.2.1 (Partition)

A partition P for the interval $[a, b]$ is a finite increasing sequence of numbers of the form

$$a = t_0 < t_1 < t_2 < \cdots < t_{i-1} < t_i < \cdots < t_{n-1} < t_n = b$$

Note: A partition subdivides the interval $[a, b]$ into n subintervals

$$[t_0, t_1], [t_1, t_2], \cdots, [t_{i-1}, t_i], \cdots, [t_{n-2}, t_{n-1}], [t_{n-1}, t_n]$$

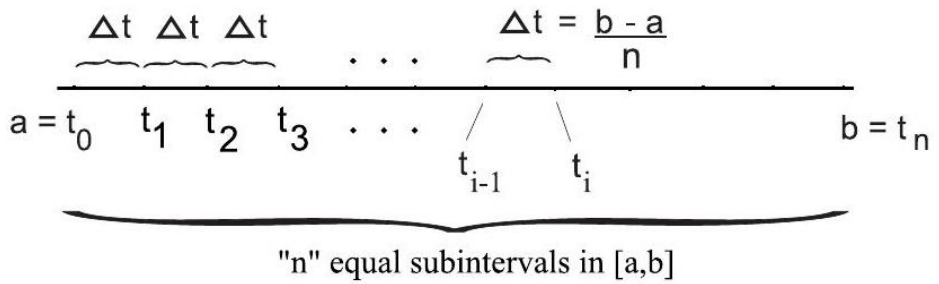
Let

$$\Delta t_i = t_i - t_{i-1}$$

and let the norm of the partition P be

$$\|P\| = \max\{\Delta t_1, \Delta t_2, \ldots, \Delta t_n\}$$

Regular n-Partition



Definition 6.2.2 (Regular n-Partition)

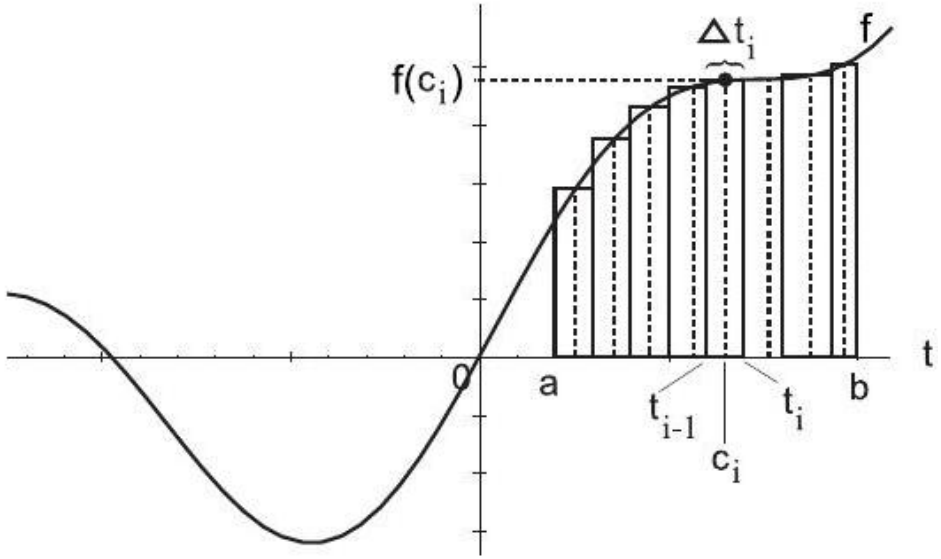
Given an interval $[\boldsymbol{a}, \boldsymbol{b}]$ and an $\boldsymbol{n} \in \mathbb{N}$, the regular $\boldsymbol{n}$-partition of $[a, b]$ is the partition $P^{(n)}$ with

$$a = t_0 < t_1 < t_2 < \cdots < t_{i-1} < t_i < \cdots < t_{n-1} < t_n = b$$

of $[a, b]$ where each subinterval has the same length $\Delta t_i = \frac{b-a}{n}$. Hence,

$$t_i = a + i \cdot \frac{b-a}{n}$$

Introduction to Riemann Sums



Definition 6.2.3 (Riemann Sum)

Given a bounded function f on $[\boldsymbol{a}, \boldsymbol{b}]$, a partition P

$$a = t_0 < t_1 < t_2 < \cdots < t_{i-1} < t_i < \cdots < t_{n-1} < t_n = b$$

of $[a, b]$, and a set $\{c_1, c_2, \ldots, c_n\}$ where $c_i \in [t_{i-1}, t_i]$, then a Riemann sum for $\boldsymbol{f}$ with respect to P is a sum of the form

$$S = \sum_{i=1}^{n} f(c_i)\, \Delta t_i$$

Definition 6.2.4 (Riemann Integral)

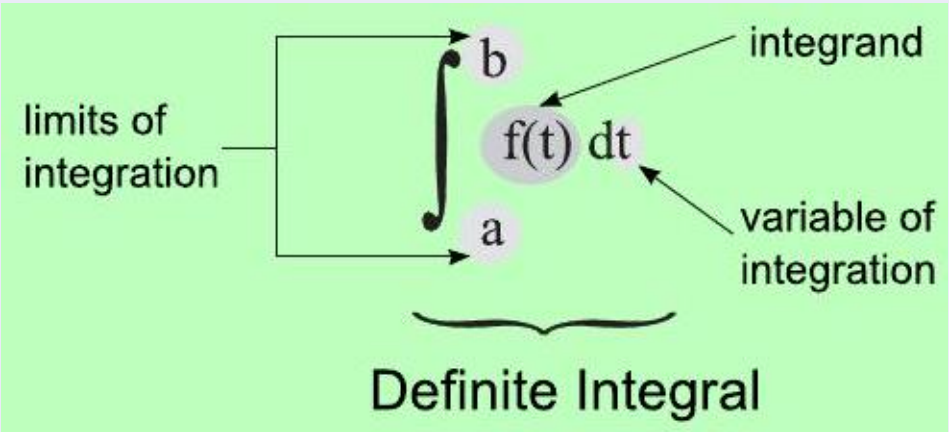
We say that a bounded function $\boldsymbol{f}$ is integrable on $[\boldsymbol{a}, \boldsymbol{b}]$ if there exists a unique number $I \in \mathbb{R}$ such that if whenever $\{P_n\}$ is a sequence of partitions with $\lim_{n\to\infty} \|P_n\| = \mathbf{0}$ and $\{S_n\}$ is any sequence of Riemann sums associated with the P_n's, we have

$$\lim_{n\to\infty} S_n = I$$

In this case, we call $\boldsymbol{I}$ the integral of $\boldsymbol{f}$ over $[\boldsymbol{a}, \boldsymbol{b}]$ and denote it by

$$\int_a^b f(t)dt$$

The points a and b are called the limits of integration and the function $f(t)$ is called the integrand. The variable t is called the variable of integration.



Definition of the Integral

Note: The variable of integration is sometimes called a dummy variable in the sense that if we were to replace t's by x's everywhere, we would not change the value of the integral.

Question: Are all bounded functions on $[a, b]$ integrable?

Answer: No. Let

$$f(x) = \begin{cases} 1 & \text{if } x \in \mathbb{Q} \\ -1 & \text{if } x \notin \mathbb{Q} \end{cases}$$

Then f is not integrable on any interval $[\boldsymbol{a}, \boldsymbol{b}]$.

The Integrability Theorem

Theorem 6.2.5 (The Integrability Theorem for Continuous Functions)

Let f be continuous on $[a, b]$. Then f is integrable on $[a, b]$. Moreover,

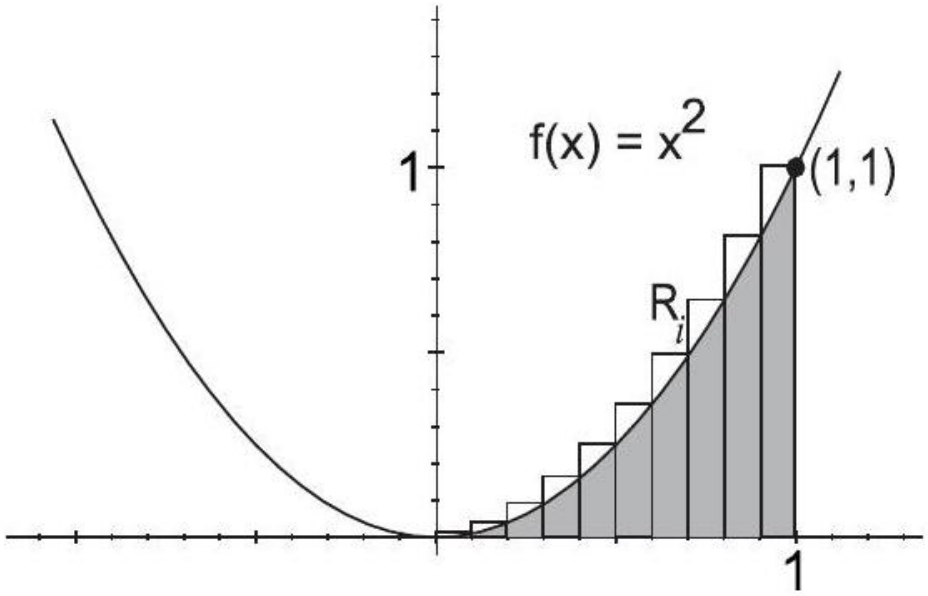
$$\int_a^b f(t)dt = \lim_{n\to\infty} S_n$$

where

$$S_n = \sum_{i=1}^{n} f(c_i)\,\Delta t_i$$

is any Riemann sum associated with the regular n-partitions.

The Integrability Theorem



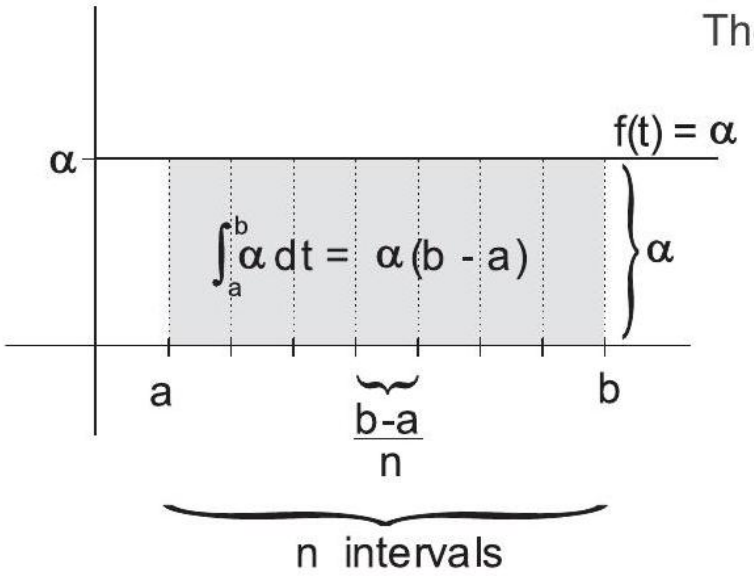
Example: Since

$$R_n = \frac{1}{n^3}\sum_{i=1}^{n} i^2 = \frac{2+\frac{3}{n}+\frac{1}{n^2}}{6}$$

we get

$$\int_0^1 x^2 dx = \lim_{n\to\infty} \frac{2+\frac{3}{n}+\frac{1}{n^2}}{6} = \frac{1}{3}$$

Integrating Constant Functions



Example:

Let $f(t) = \alpha$ for each $t \in [a, b]$.
Then

$$\begin{aligned} R_n &= \sum_{i=1}^{n} f(t_i)\Delta t_i \\ &= \sum_{i=1}^{n} \alpha\left(\frac{b-a}{n}\right) \\ &= \alpha \sum_{i=1}^{n} \frac{b-a}{n} \\ &= \alpha(b-a) \end{aligned}$$

Since $R_n = \alpha(b-a)$ for each n,

$$\int_a^b \alpha dt = \alpha(b-a)$$

Properties of Riemann Integrals

Theorem 6.2.6

Assume that f and g are integrable on the interval $[a, b]$. Then

i) For any $c \in \mathbb{R}$, $\int_a^b cf(t)dt = c\int_a^b f(t)dt$

ii) $\int_a^b (f+g)(t)dt = \int_a^b f(t)dt + \int_a^b g(t)dt$

iii) If $m \leq f(t) \leq M$ for all $t \in [a, b]$, then

$$m(b-a) \leq \int_a^b f(t)dt \leq M(b-a)$$

iv) If $0 \leq f(t)$ for all $t \in [a, b]$, then $0 \leq \int_a^b f(t)dt$.

v) If $g(t) \leq f(t)$ for all $t \in [a, b]$, then $\int_a^b g(t)dt \leq \int_a^b f(t)dt$.

vi) The function $|f|$ is integrable on $[a, b]$ and

$$\left|\int_a^b f(t)dt\right| \leq \int_a^b |f(t)|dt$$

Properties of Riemann Integrals

Proof of (iii): Assume that

$$m \leq f(t) \leq M$$

for all $t \in [a, b]$. Let

$$a = t_0 < t_1 < t_2 < \cdots < t_{i-1} < t_i < \cdots < t_{n-1} < t_n = b$$

be any partition of $[a, b]$. Then

$$m(b-a) = \sum_{i=1}^{n} m\Delta t_i \leq \sum_{i=1}^{n} f\left(t_i\right) \Delta t_i \leq \sum_{i=1}^{n} M\Delta t_i = M(b-a),$$

since

$$\sum_{i=1}^{n} \Delta t_i = b - a.$$

Properties of Riemann Integrals

Remark: So far in defining the Riemann Integral we have always considered integrals of the form

$$\int_a^b f(t)dt$$

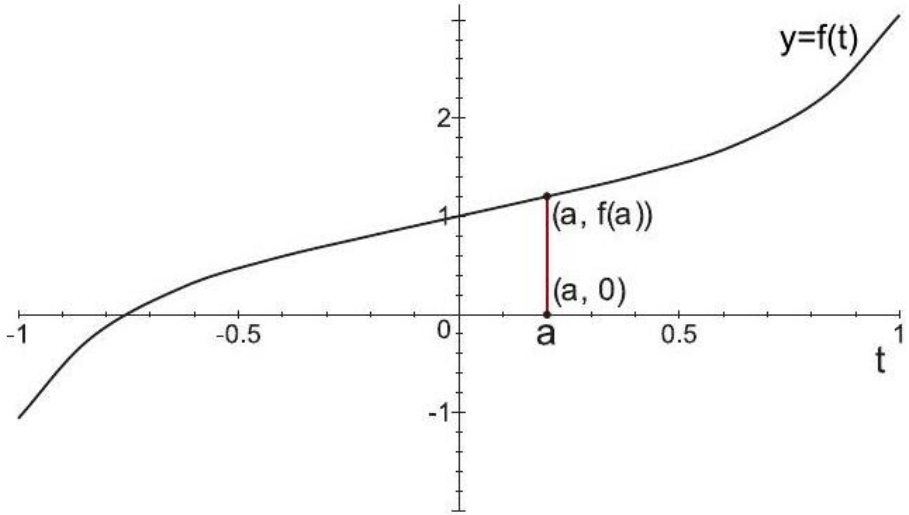
where $\boldsymbol{a < b}$. However, it is necessary to give meaning to

$$\int_a^a f(t)dt$$

and to

$$\int_b^a f(t)dt$$

Properties of Riemann Integrals



Question: How should we define $\int_a^a f(t)dt$?

Remark: We can see that the line segment has height $\boldsymbol{f(a)}$ but length $\mathbf{0}$. As such it makes sense to define its "area" to be $\mathbf{0}$.

Definition 6.2.7 (Identical Limits of Integration: $\int_a^a f(t)dt$)

Let $f(t)$ be defined at $t = a$. Then we define

$$\int_a^a f(t)dt = 0$$

Properties of Riemann Integrals

Remark: In the definition of

$$\int_a^b f(t)dt$$

where $a < b$, we began at the left-hand endpoint a of an interval $[a, b]$ and moved to the right towards b. In the case of the integral

$$\int_b^a f(t)dt$$

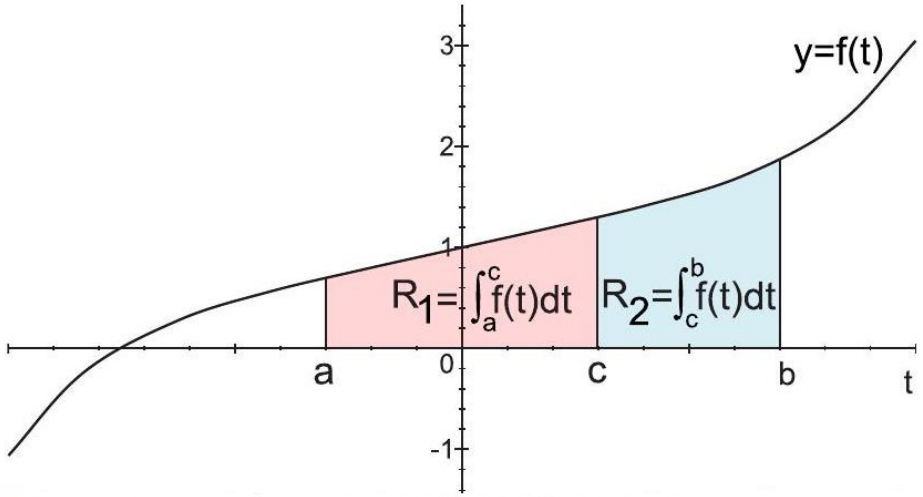
where $a < b$, we are suggesting that using the interval $[a, b]$ we move from b to the left towards a. This is the opposite or negative of the original orientation.

Definition 6.2.8 (Switching the Limits of Integration)

Let f be integrable on the interval $[a, b]$ where $a < b$. Then we define

$$\int_b^a f(t)dt = -\int_a^b f(t)dt$$

Properties of Riemann Integrals



Remark: Assume that $\boldsymbol{f}$ is continuous and positive on $[\boldsymbol{a}, \boldsymbol{b}]$ with $a < c < b$. Since

$$R = R_1 + R_2$$

we should have

$$\int_a^b f(t)dt = \int_a^c f(t)dt + \int_c^b f(t)dt$$

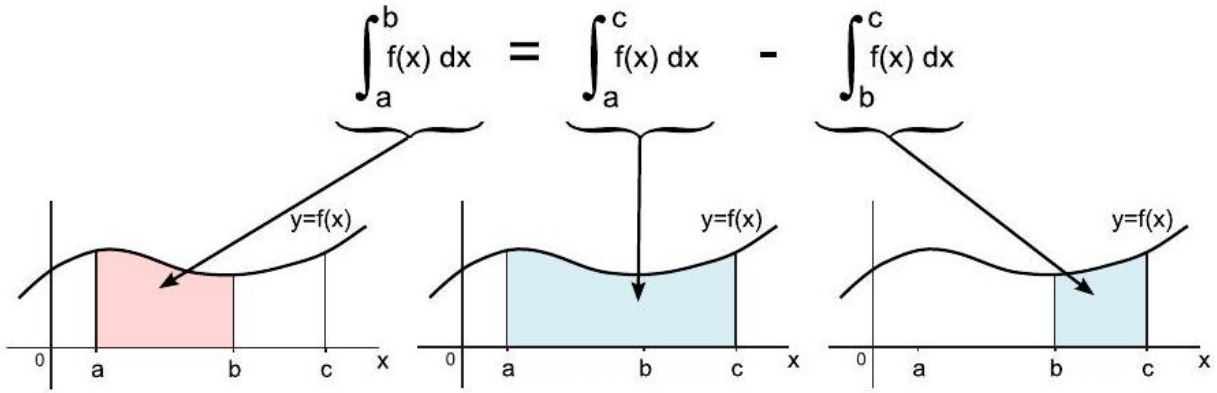
Properties of Riemann Integrals

Theorem 6.2.9 (Integrals over Subintervals)

Assume that f is integrable on an interval I containing a, b and c. Then

$$\int_a^b f(t)dt = \int_a^c f(t)dt + \int_c^b f(t)dt$$

Properties of Riemann Integrals



Remark: Assume that f is integrable on the interval $[a, c]$ where $a < b < c$. Since $\int_b^c f(x)dx = -\int_c^b f(x)dx$, we get

$$\begin{aligned}\int_a^b f(x)dx &= \int_a^c f(x)dx - \int_b^c f(x)dx \\ &= \int_a^c f(x)dx + \int_c^b f(x)dx\end{aligned}$$

Properties of Riemann Integrals

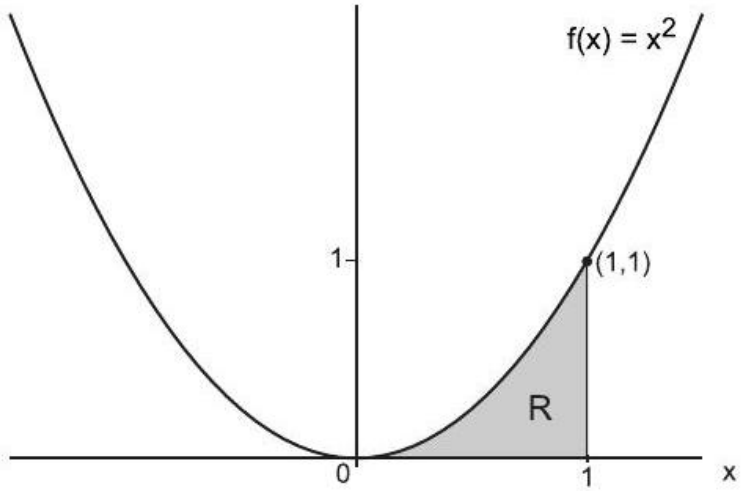
Theorem 6.2.10 (Mean Value Theorem for Integrals)

Assume that f is continuous on $[a, b]$. Then there exists $a \leq c \leq b$ such that

$$f(c) = \frac{1}{b-a} \int_a^b f(t)dt.$$

Section 3

Interpretations of Riemann Integral



Remark: We have already seen that the area of the region R bounded by the graph of $f(x) = x^2$, by the x-axis, and by the lines $x = 0$ and $x = 1$ is

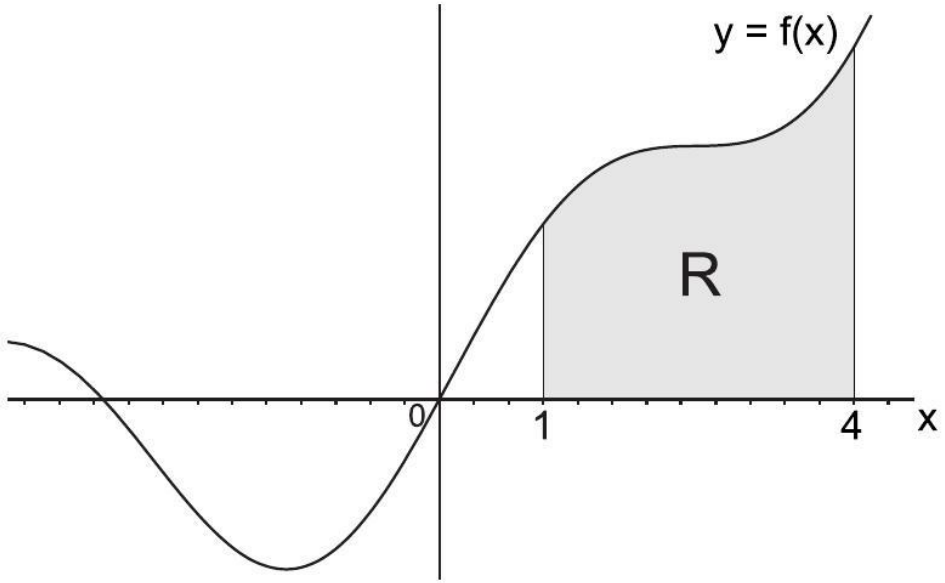
$$\int_0^1 x^2 dx$$

In fact, whenever $f(x) \geq \mathbf{0}$ on all of $[\boldsymbol{a}, \boldsymbol{b}]$, the area under the graph of f and above the x-axis bounded by the lines $x = a$ and $x = b$ will be

$$\int_a^b f(x) dx$$

Question: What happens if $f(x) \leq 0$ on some part of $[a, b]$?

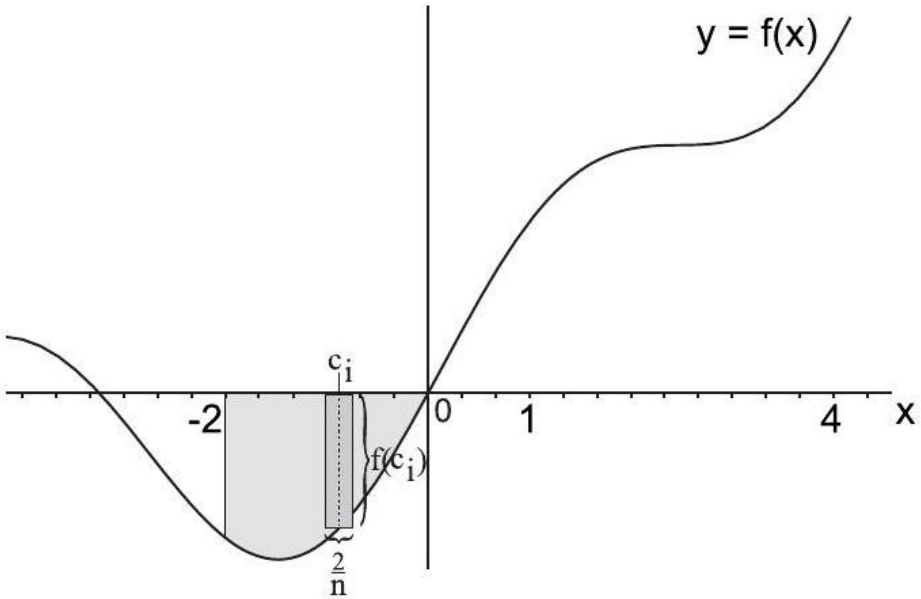
Geometric Interpretation of Riemann Integral



Example: Assume that f is as shown in the diagram. Then $\int_1^4 f(x)dx$ is the area of the region R bounded by the graph of f, the x-axis, and the lines $x = 1$ and $x = 4$.

Question: Suppose instead that we wanted to calculate $\int_{-2}^0 f(x)dx$. What would this represent?

Geometric Interpretation of Riemann Integral



Observation: Consider a term

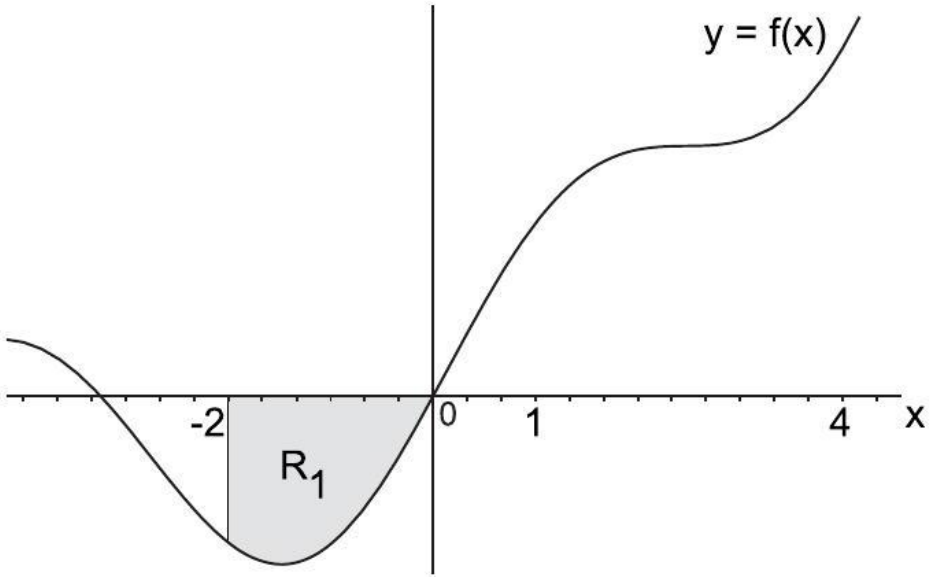
$$f\left(c_i\right)\frac{2}{n}$$

in a generic Riemann sum. Since $f\left(c_i\right)$ is the negative of the height of the pictured rectangle

$$S_n = \sum_{i=1}^{n} f\left(c_i\right)\frac{2}{n}$$

approximates the negative of the area bounded by the graph of f, the x-axis, and the lines $x = -2$ and $x = 0$.

Geometric Interpretation of Riemann Integral



Observation (continued): Since

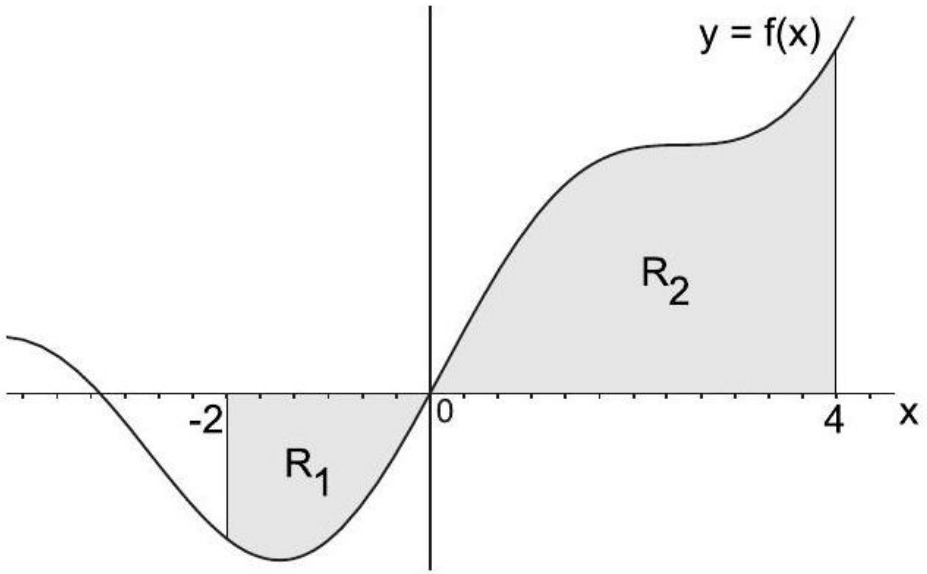
$$S_n = \sum_{i=1}^{n} f(c_i) \frac{2}{n}$$

approximates the negative of the area bounded by the graph of f, the x-axis, and the lines $x = -2$ and $x = 0$,

$$\int_{-2}^{0} f(x)dx = \lim_{n\to\infty} S_n = \lim_{n\to\infty} \sum_{i=1}^{n} f(c_i) \frac{2}{n}$$

is the negative of the area of the region R_1.

Geometric Interpretation of Riemann Integral



We have

$$\int_{-2}^{4} f(x)dx = \int_{-2}^{0} f(x)dx + \int_{0}^{4} f(x)dx$$
$$= R_2 - R_1$$

Geometric Interpretation of Riemann Integral

Geometric Interpretation of Riemann Integral

If f is a continuous function on the interval $[a, b]$, then

$$\int_a^b f(x)dx$$

represents the area of the region under the graph of f that lies above the x-axis between $x = a$ and $x = b$ minus the area of the region above the graph of f that lies below the x-axis between $x = a$ and $x = b$.

Geometric Interpretation of Riemann Integral

Example 6.3.1

Evaluate

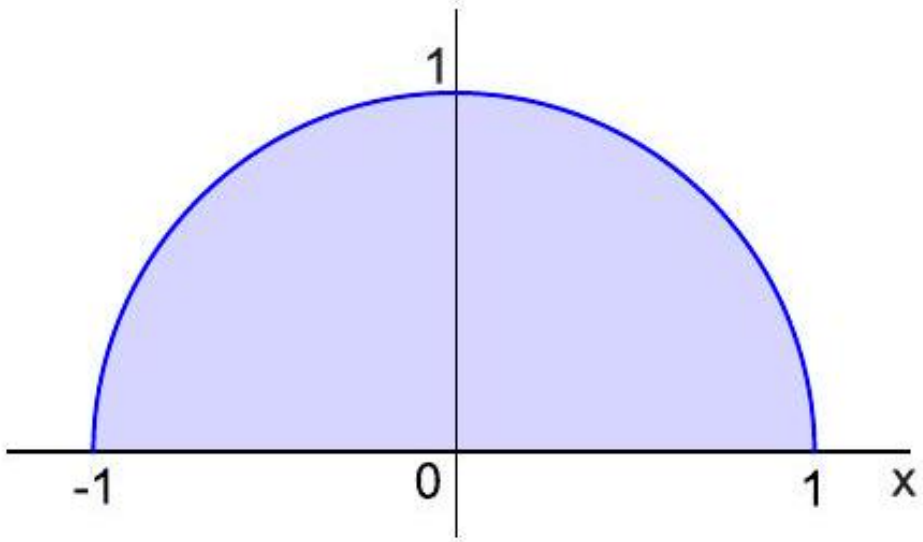
$$\int_{-2}^{3}(2x-1)dx$$

geometrically as the difference in the area of two triangles.

Solution:

$$\begin{aligned}\int_{-2}^{3}(2x-1)dx &= \int_{-2}^{\frac{1}{2}}(2x-1)dx + \int_{\frac{1}{2}}^{3}(2x-1)dx \\ &= -R_1 + R_2 \\ &= -\frac{5\times\left(\frac{1}{2}-(-2)\right)}{2} + \frac{5\times\left(3-\frac{1}{2}\right)}{2} \\ &= -\frac{25}{4} + \frac{25}{4} = 0\end{aligned}$$

Geometric Interpretation of Riemann Integral



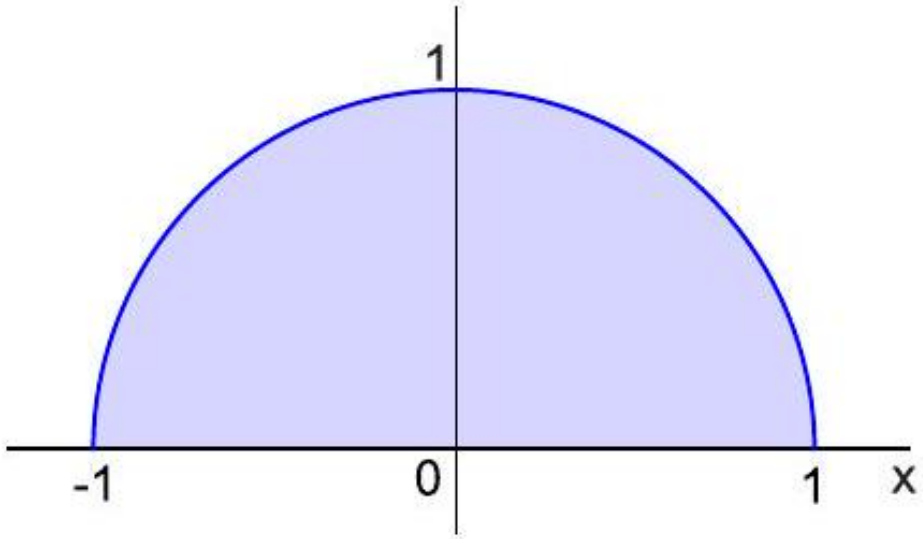
Example: Evaluate

$$\int_{-1}^{1} \sqrt{1-x^2}dx$$

Solution: The shape of this region is a semi-circle with radius 1 . It follows that

$$\int_{-1}^{1} \sqrt{1-x^2}dx = \frac{\pi}{2}$$

Geometric Interpretation of Riemann Integral



Example: Evaluate

$$\int_{-1}^{1} \sqrt{1-x^2}dx$$

Solution: The shape of this region is a semi-circle with radius 1 . It follows that

$$\int_{-1}^{1} \sqrt{1-x^2}dx = \frac{\pi}{2}$$

Question: How would we evaluate

$$\int_{-1}^{\frac{1}{2}} \sqrt{1-x^2}dx$$

Section 4

The Fundamental Theorem of Calculus

Integral Functions

Remark: Assume that the function f is continuous on an interval $[a, b]$. Define the integral function G on $[a, b]$ by

$$G(x) = \int_a^x f(t)dt$$

Questions: What are the properties of G ?

Is it continuous?

Is it differentiable?

If so what is $G'(x)$?

Fundamental Theorem of Calculus (Part 1)

Theorem 6.4.1 (Fundamental Theorem of Calculus (Part 1) (FTC1))

Assume that f *is continuous on an open interval* I *containing a point* a*. Let*

$$G(x) = \int_a^x f(t)dt$$

Then $G(x)$ *is differentiable at each* $x \in I$ *and*

$$G'(x) = f(x)$$

Equivalently,

$$G'(x) = \frac{d}{dx}\int_a^x f(t)dt = f(x)$$

Fundamental Theorem of Calculus (Part 1)

Proof of the FTC1: Assume that

$$G(x) = \int_a^x f(t)dt$$

and that $\boldsymbol{f}$ is continuous at $\boldsymbol{x_0} \in \boldsymbol{I}$. Let $\epsilon > \mathbf{0}$. Then there exists a $\delta > 0$ so that if $0 < |c - x_0| < \delta$, then

$$|f(c) - f(x_0)| < \epsilon$$

If $0 < |x - x_0| < \delta$, then

$$\begin{aligned} \frac{G(x) - G(x_0)}{x - x_0} &= \frac{\int_a^x f(t)dt - \int_a^{x_0} f(t)dt}{x - x_0} \\ &= \frac{1}{x - x_0} \int_{x_0}^x f(t)dt \end{aligned}$$

By the Average Value Theorem, there exists a c between $\boldsymbol{x}$ and x_0 with

$$f(c) = \frac{1}{x - x_0} \int_{x_0}^x f(t)dt$$

Fundamental Theorem of Calculus (Part 1)

Proof of the FTC1 (continued):

If $0 < |x - x_0| < \delta$, then since $0 < |c - x_0| < \delta$,

$$\left| \frac{G(x) - G(x_0)}{x - x_0} - f(x_0) \right| = |f(c) - f(x_0)| < \epsilon$$

By the definition of a limit we get that

$$\begin{aligned} G'(x_0) &= \lim_{x \to x_0} \frac{G(x) - G(x_0)}{x - x_0} \\ &= f(x_0) \end{aligned}$$

Fundamental Theorem of Calculus (Part 1)

Example: Find $\boldsymbol{F}'(x)$ if

$$F(x) = \int_3^x e^{t^2} dt$$

Solution: Since $f(t) = e^{t^2}$ is a continuous function, the Fundamental Theorem of Calculus 1 tells us we can replace t by x in $\boldsymbol{f}(\boldsymbol{t})$ to get

$$F'(x) = e^{x^2}$$

Fundamental Theorem of Calculus (Part 2)

Example: Evaluate

$$\int_0^2 t^3 dt$$

Observation: If

$$G(x) = \int_0^x t^3 dt$$

then

$$G(2) = \int_0^2 t^3 dt$$

We know from the FTC1 that

$$G'(x) = x^3$$

so G is an antiderivative of x^3. Hence there exists a constant C such that

$$G(x) = \int_0^x t^3 dt = \frac{x^4}{4} + C$$

Question: How does this help us?

Fundamental Theorem of Calculus (Part 2)

Note: We have just seen that

$$G(x) = \int_0^x t^3 dt = \frac{x^4}{4} + C$$

for some constant $C \in \mathbb{R}$.

However, we also know that

$$0 = \int_0^0 t^3 dt = G(0) = \frac{0^4}{4} + C = C$$

so

$$G(x) = \int_0^x t^3 dt = \frac{x^4}{4}$$

Finally,

$$\int_0^2 t^3 dt = G(2) = \frac{2^4}{4} = 4$$

Question: Did we really need to find C ?

Fundamental Theorem of Calculus (Part 2)

Key Observation:

Let $\boldsymbol{F}$ and G be any two antiderivatives of the same function f. Then

$$G(x) = F(x) + C$$

Let $a, b \in \mathbb{R}$. Then

$$\begin{aligned} G(b) - G(a) &= (F(b) + C) - (F(a) + C) \\ &= F(b) - F(a) \end{aligned}$$

Fundamental Theorem of Calculus (Part 2)

Key Observation (continued):

Assume that f is continuous and that we want to know

$$\int_a^b f(t)dt$$

Let

$$G(x) = \int_a^x f(t)dt$$

Then G is an antiderivative of f and if $\boldsymbol{F}$ is any other antiderivative of f, then

$$\begin{aligned}\int_a^b f(t)dt &= G(b) \\ &= G(b) - G(a) \quad (\text{ since } G(a) = 0) \\ &= F(b) - F(a)\end{aligned}$$

Fundamental Theorem of Calculus (Part 2)

Example: Evaluate

$$\int_0^2 t^3 dt$$

Solution: We know that

$$F(x) = \frac{x^4}{4}$$

is an antiderivative for $f(x) = x^3$.

Hence

$$\begin{aligned}\int_0^2 t^3 dt &= F(2) - F(0) \\ &= \frac{2^4}{4} - \frac{0^4}{4} \\ &= 4\end{aligned}$$

Fundamental Theorem of Calculus (Part 2)

Theorem 6.4.2 (FTC2: Fundamental Theorem of Calculus (Part 2))

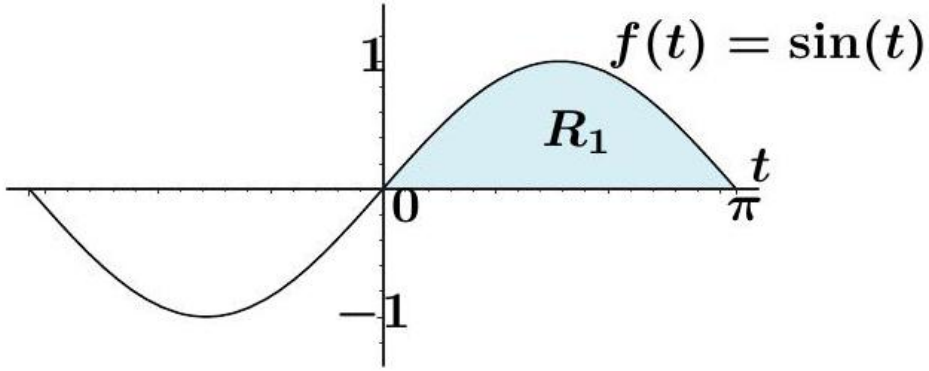
Assume that f is continuous and that $\boldsymbol{F}$ is any antiderivative of f. Then

$$\int_a^b f(t)dt = F(b) - F(a).$$

Notation: We write

$$F(x)\Big|_a^b = F(b) - F(a)$$

Fundamental Theorem of Calculus (Part 2)



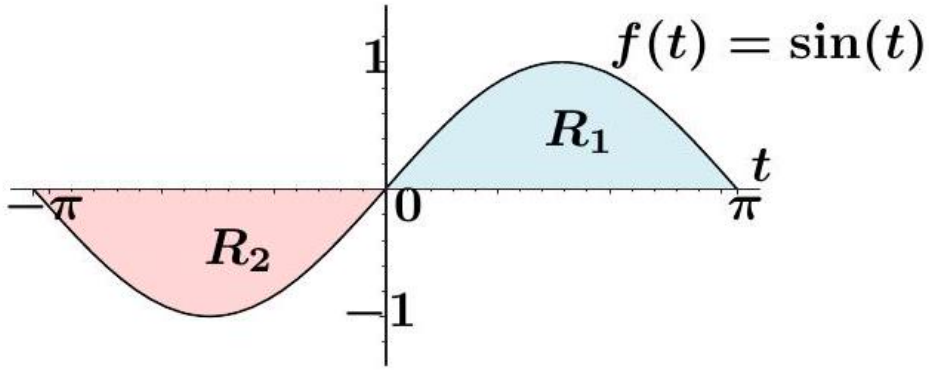
Example: Evaluate

$$\int_0^{\pi} \sin(t)dt$$

Solution: Since $f(t) = \sin(t)$ is continuous and $F(t) = -\cos(t)$ is an antiderivative of f, by the FTC2, we have

$$\begin{aligned}\int_0^{\pi} \sin(t)dt &= -\cos(t)|_0^{\pi} \\ &= -\cos(\pi) - (-\cos(0)) \\ &= 2\end{aligned}$$

Fundamental Theorem of Calculus (Part 2)



Example: Evaluate

$$\int_{-\pi}^{\pi} \sin(t)dt$$

Solution: Using a geometrical argument, this integral should equal $\mathbf{0}$.

$$\begin{aligned}\int_{-\pi}^{\pi} \sin(t)dt &= -\cos(t)|_{-\pi}^{\pi} \\ &= -\cos(\pi) - (-\cos(-\pi)) \\ &= 0\end{aligned}$$

Fundamental Theorem of Calculus (Part 2)

Key Remark: It is important that we emphasize the difference between the meaning of

$$\int_a^b f(t)dt \quad \text{and} \quad \int f(t)dt$$

The first expression, the Riemann integral

$$\int_a^b f(t)dt$$

is also called a **definite integral**. It represents a number that is defined as a limit of Riemann sums.

The second expression,

$$\int f(t)dt$$

is called an **indefinite integral**. It represents the family of all functions that are antiderivatives of the given function f.

Section 5

Integration by Parts and Change of Variables

Theorem 6.5.1 (Integration by parts, shorted as IBP)

Let $f, g : [a, b] \to \mathbb{R}$ be integrable such that everything below exists. Then

$$\int_a^b f(x)g'(x)\,dx = f(b)g(b) - f(a)g(a) - \int_a^b f'(x)g(x)\,dx.$$

Proof.

By the fundamental theorem of calculus,

$$\int_a^b (f(x)g'(x) + f'(x)g(x))\,dx = \int_a^b (fg)'(x)\,dx = f(b)g(b) - f(a)g(a).$$

The result follows after rearrangement. □

Remark: If we write $f'(x)dx = d(f(x))$ and $g'(x)dx = d(g(x))$, we can write the IBP as

$$\int_a^b f(x)d(g(x)) = f(b)g(b) - f(a)g(a) - \int_a^b g(x)\,d(f(x)).$$

Example 6.5.2

Compute $\int_0^x x^2 \cos x dx$.

Solution. By integration by parts,

$$\begin{aligned}\int_0^\pi x^2 \cos x dx &= \int_0^\pi x^2 d(\sin x) = x^2 \sin x\big|_0^\pi - \int_0^\pi \sin x d\left(x^2\right) \\ &= -2\int_0^\pi x \sin x dx = 2\int_0^\pi x d(\cos x) \\ &= 2x\cos x\big|_0^\pi - 2\int_0^\pi \cos x dx \\ &= -2\pi - 2\sin x\big|_0^\pi = -2\pi.\end{aligned}$$

Theorem 6.5.3 (Integration by substitution)

Let $g : [a, b] \to I$ be a differentiable function with a continuous derivative, where $I \subset \mathbb{R}$ is an interval. Suppose that $f : I \to \mathbb{R}$ is a continuous function. Then:

$$\int_a^b f(g(x)) \cdot g'(x)dx = \int_{g(a)}^{g(b)} f(u)du.$$

Proof.

Since f is continuous, it has an antiderivative F. The composite function $F \circ g$ is then defined. Since g is differentiable, combining the chain rule and the definition of an antiderivative gives:

$$(F \circ g)'(x) = F'(g(x)) \cdot g'(x) = f(g(x)) \cdot g'(x)$$

Applying the fundamental theorem of calculus twice gives:

$$\begin{aligned}\int_a^b f(g(x)) \cdot g'(x)dx &= \int_a^b (F \circ g)'(x)dx \\ &= (F \circ g)(b) - (F \circ g)(a) \\ &= F(g(b)) - F(g(a)) \\ &= \int_{g(a)}^{g(b)} f(u)du.\end{aligned}$$

□

Remark: We can think of "integration by parts" as what you get by integrating the product rule, and "integration by substitution" as what you get by integrating the chain rule.

Example 6.5.4

Consider the integral:

$$\int_0^2 \frac{x}{\sqrt{x^2+1}} dx.$$

Make the substitution $u = x^2 + 1$ to obtain $du = 2xdx$, meaning $xdx = \frac{1}{2}du$. Therefore:

$$\begin{aligned}\int_{x=0}^{x=2} \frac{x}{\sqrt{x^2+1}} dx &= \frac{1}{2}\int_{u=1}^{u=5} \frac{du}{\sqrt{u}} \\ &= \frac{1}{2}(2\sqrt{5} - 2\sqrt{1}) \\ &= \sqrt{5} - 1.\end{aligned}$$

Example 6.5.5

For the integral

$$\int_0^1 \sqrt{1-x^2}dx$$

a variation of the above procedure is needed. The substitution $x = \sin u$ implying $dx = \cos u du$ is useful because $\sqrt{1-\sin^2 u} = \cos u$. We thus have:

$$\begin{aligned}
\int_0^1 \sqrt{1-x^2}dx &= \int_0^{\pi/2} \sqrt{1-\sin^2 u}\cos u du \\
&= \int_0^{\pi/2} \cos^2 u du \\
&= \left[\frac{u}{2} + \frac{\sin(2u)}{4}\right]_0^{\pi/2} \\
&= \frac{\pi}{4} + 0 = \frac{\pi}{4}
\end{aligned}$$

Section 6

Improper Integrals

In this section, we extend the definition of integrals to functions either of which domain is an infinite interval or with infinite discontinuity in [a,b]. In either case, the integral is called an improper integral.

Definition 6.6.1 (Continuous integrands on unbounded domains)

Let f be a function and $a, b \in \mathbb{R}$.
(1) When $\int_a^t f(x)dx$ exists for all $t \geq a$ and has a limit as $t \to \infty$, define

$$\int_a^{\infty} f(x)dx = \lim_{t\to\infty} \int_a^t f(x)dx$$

(2) When $\int_s^b f(x)dx$ exists for all $s \leq b$ and has a limit as $s \to -\infty$, define

$$\int_{-\infty}^b f(x)dx = \lim_{s\to-\infty} \int_s^b f(x)dx$$

(3) When $\int_a^{\infty} f(x)dx$ and $\int_{-\infty}^a f(x)dx$ exist for some $a \in \mathbb{R}$, define

$$\int_{-\infty}^{\infty} f(x)dx = \int_{-\infty}^a f(x)dx + \int_a^{\infty} f(x)dx$$

Remarks

- $\int_a^{\infty} f(x)dx$ and $\int_{-\infty}^{b} f(x)dx$ are called convergent if the corresponding limits exist and called divergent otherwise.
- If $\lim_{t\to\infty} \int_a^t f(x)dx = \pm\infty$, we write $\int_a^{\infty} f(x)dx = \pm\infty$.
- If $\lim_{s\to-\infty} \int_s^b f(x)dx = \pm\infty$, we write $\int_{-\infty}^{b} f(x)dx = \pm\infty$.
- If $f \geq 0$ on $[a, \infty)$, then the area of the region

$$S = \{(x, y) \mid x \geq a, 0 \leq y \leq f(x)\}$$

is defined to be $\int_a^{\infty} f(x)dx$.

Example 6.6.2

To compute $\int_{-\infty}^{0} xe^x dx$, note that

$$\int xe^x dx = (x-1)e^x + C.$$

Then,

$$\lim_{t\to-\infty} \int_t^0 xe^x dx = \lim_{t\to-\infty} \left[(1-t)e^t - 1\right] = -1.$$

Example 6.6.3

Consider the integral $\int_{-\infty}^{\infty} \frac{1}{1+x^2}dx$. Note that

$$\int_{-\infty}^{0} \frac{1}{1+x^2}dx = \lim_{t\to-\infty} \int_{t}^{0} \frac{1}{1+x^2}dx = \lim_{t\to-\infty} \left(-\tan^{-1} t\right) = \frac{\pi}{2}$$

and

$$\int_{0}^{\infty} \frac{1}{1+x^2}dx = \lim_{t\to\infty} \int_{0}^{t} \frac{1}{1+x^2}dx = \lim_{t\to\infty} \tan^{-1} t = \frac{\pi}{2}$$

This implies $\int_{-\infty}^{\infty} \frac{1}{1+x^2}dx = \pi$.

Example 6.6.4

Consider the integral $\int_1^\infty x^p dx$. Note that

$$\int x^p dx = \frac{x^{p+1}}{p+1} + C \quad \text{for } p \neq -1, \qquad \int x^{-1} dx = \ln x + C.$$

This implies

$$\lim_{t\to\infty} \int_1^t x^p dx = \lim_{t\to\infty} \begin{cases} \frac{1}{p+1}\left(t^{p+1} - 1\right) & \text{for } p \neq -1 \\ \ln t & \text{for } p = -1 \end{cases}$$

$$= \begin{cases} -1/(1+p) & \text{for } p < -1 \\ \infty & \text{for } p \geq -1 \end{cases}.$$

Definition 6.6.5 (Discontinuous integrands)

Let f be a function.
(1) If f is continuous on $[a, b)$ but discontinuous at b, define

$$\int_a^b f(x)dx = \lim_{t \to b^-} \int_a^t f(x)dx$$

(2) If f is continuous on $(a, b]$ but discontinuous at a, define

$$\int_a^b f(x)dx = \lim_{t \to a^+} \int_t^b f(x)dx.$$

In (1) and (2), the improper integral is called convergent if the limit exists and divergent otherwise. Write $\int_a^b f(x)dx = \pm\infty$ if the limit diverges to $\pm\infty$.
(3) If f has discontinuity at $c \in (a, b)$ and $\int_a^c f(x)dx$ and $\int_c^b f(x)dx$ are convergent, define

$$\int_a^b f(x)dx = \int_a^c f(x)dx + \int_c^b f(x)dx.$$

Example 6.6.6

To evaluate $\int_2^5 \frac{1}{\sqrt{x-2}}dx$, since $\frac{1}{\sqrt{x-2}}$ is continuous on $(2,5]$ and

$$\int \frac{1}{\sqrt{x-2}}dx = 2\sqrt{x-2}+C,$$

we have

$$\int_2^5 \frac{1}{\sqrt{x-2}}dx = \lim_{t\to 2^+}\int_t^5 \frac{dx}{\sqrt{x-2}}dx = 2\lim_{t\to 2^+}(\sqrt{3}-\sqrt{t-2}) = 2\sqrt{3}.$$

Example 6.6.7

To compute $\int_0^1 \ln x dx$, note that $\ln x$ is continuous on $(0, 1]$ and

$$\int \ln x dx = x(\ln x - 1) + C.$$

This implies

$$\int_0^1 \ln x dx = \lim_{t \to 0^+} \int_t^1 \ln x dx = \lim_{t \to 0^+} (-1 - t \ln t + t) = -1.$$

Example 6.6.8

Consider $\int_0^3 \frac{dx}{x-1}$. Clearly, $\frac{1}{x-1}$ has a discontinuity at 1. Note that

$$\int_1^3 \frac{dx}{x-1} = \lim_{t\to 1^+} (\ln 2 - \ln(t-1)) = \infty,$$
$$\int_0^1 \frac{dx}{x-1} = \lim_{t\to 1^-} \ln|t-1| = -\infty.$$

This implies that $\int_0^3 \frac{dx}{x-1}$ does not exist.

Theorem 6.6.9 (Comparison test)

Let f and g be functions continuous on $[a, \infty)$. Assume that $0 \leq f \leq g$.
(1) If $\int_a^\infty g(x)dx$ is convergent, then $\int_a^\infty f(x)dx$ is convergent.
(2) If $\int_a^\infty f(x)dx$ is divergent, then $\int_a^\infty g(x)dx$ is divergent.

Example 6.6.10

Consider the integral $\int_0^\infty e^{-x^2}dx$. Note that

$$e^{-x^2} \leq e^{-x} \quad \forall x \geq 1.$$

By the comparison test, $\int_0^\infty e^{-x^2}dx$ is convergent, since

$$\begin{aligned}\int_0^\infty e^{-x^2}dx &= \int_0^1 e^{-x^2}dx + \int_1^\infty e^{-x^2}dx \\ &\leq \int_0^1 e^{-x^2}dx + \int_1^\infty e^{-x}dx < \infty.\end{aligned}$$

Example 6.6.11

Consider the integral $\int_0^\infty xe^x dx$. Note that

$$xe^x \geq x \quad \forall x \geq 0.$$

By the comparison test, $\int_0^\infty xe^x dx$ is divergent.